

Patrick Smith

Building production platforms from scratch: APIs, databases, cloud infrastructure, and everything between

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📄 Summary

Full-stack systems engineer with 13 years building production platforms from the ground up. I design and implement complete systems—backend APIs, databases, cloud infrastructure, and deployment pipelines—translating abstract business needs into reliable software that ships and runs.

At Home Remodel Buddy, I architected four integrated production systems processing thousands of daily transactions with sub-250ms response times. From optimizing MySQL queries (hours → minutes) to responding to security incidents (WordPress RAT recovered in 4 hours with zero data loss via memory forensics), I own systems across their entire lifecycle.

I specialize in integration architecture—connecting payment processors, CRM systems, and third-party APIs where the challenge is making documented behavior match actual behavior. My approach: build systems that are straightforward to debug, deploy without anxiety, and run reliably enough that everyone else can get back to their work.

🧰 Experience

Founding Engineer

May 2011 – Sept 2024

Home Remodel Buddy, San Francisco, CA

- Designed and maintained four integrated production systems for 13 years: API gateway processing thousands of daily requests (sub-250ms response times), wholesale distribution platform with scoring algorithm balancing return rates and pricing, customer portal serving 740+ contractors via Knockout.js SPA, and Twilio-based IVR validation system.
- Built REST and SOAP APIs integrating payment processors (Stripe, PayPal), CRM systems, and marketing platforms—the kind of integration work where half the challenge is making documented API behavior match actual behavior.
- Developed wholesale lead distribution scoring algorithm balancing partner historical return rates (2-25%), pricing tiers, and load distribution, generating \$25K-\$250K monthly revenue per partner while maintaining quality standards.
- Optimized MySQL geographic matching queries from hours to minutes by replacing $O(n^2)$ haversine calculations with bounding box pre-computation and indexed range queries, enabling near-real-time lead distribution across 740+ contractors.
- Managed AWS and Azure infrastructure including VPCs, virtual networks, and custom blue-green deployment system using Git pulls to versioned directories with atomic symlink switching—enabling instant rollbacks and zero-downtime deployments.
- Implemented production monitoring with Grafana and Prometheus tracking server metrics (CPU, memory, I/O), API response times, error rates, and business metrics. Automated alerting meant finding problems before customers did.
- Responded to WordPress remote access trojan by dumping process memory via `/proc/{pid}/mem`, reverse-engineering obfuscated PHP (base64/eval chains), identifying exploit vector, and completing full restoration in 4 hours with zero data loss.
- Provided 13 years of on-call support as sole technical engineer, diagnosing production incidents using Brendan Gregg's 60-second performance methodology and advanced debugging tools (strace, gdb, lsof, tcpdump, iptables)—from database deadlocks to firewall misconfigurations.

⚙️ Skills

Languages

Python JavaScript
PHP SQL
Bash

Cloud & Infrastructure

AWS Azure
Linux Docker
Git CI/CD

Databases & Data

MySQL Redis
ETL NumPy
Pandas sklearn

APIs & Integration

REST SOAP
OAuth JSON/XML
Webhooks

Monitoring & Security

Grafana Prometheus
IDS PCI
Incident Response

Networking

VPC VPN
DNS Firewalls

👥 References

Christopher
Levine, Director
of Eng

Lyft

415-827-2097,
clevine@keyobjects.com

Barry
Willingham,
CEO

*Funnel
Social*

415-655-1844,
barry@swimmingpoolquotes.com

Education

A.S Business/Economics

2005-2007

Gavilan College, Gilroy, CA

Computer Science Coursework

2003-2005

California Polytechnic University, San Luis Obispo, San Luis Obispo, CA

CS109x: Introduction to Data Science with Python

July, 2025

HarvardX (Online Learning Initiative of Harvard University), <https://courses.edx.org/certificates/af05411c855d4ab7aaf9c8d90b29bd09>

Machine learning with sklearn, data analysis with Pandas/NumPy, regression and classification models, cross-validation, regularization, hypothesis testing, and statistical modeling

PH125.3x: Data Science: Probability

July, 2025

HarvardX (Online Learning Initiative of Harvard University), <https://courses.edx.org/certificates/73a1329507e642a0977384e4344c650e>

Probability theory, random variables, Monte Carlo simulations, Central Limit Theorem, expected values, standard errors, statistical inference foundations, and sampling models

Master Rank - Linux Troubleshooting

2025

SadServers.com, <https://sadservers.com/u/patrick/>

Completed advanced Linux troubleshooting scenarios including system debugging, performance analysis, networking issues, service failures, and production incident simulation

Projects

Gridiron Dynasty

2025

Multi-agent AI system for sports simulation game development

Architected collaborative AI agent system using Claude Code achieving 84% performance improvement (720s → 75s bootstrap) and 90% memory reduction (2.27GB → 224MB) through systematic optimization, parallel processing, and config caching. Designed specialized agent roles (Director, Architect, Engineer, Reviewer) with automated code review pipelines and comprehensive test coverage (83+ test cases).

Silent Disco

2025

Local network audio broadcasting system for synchronized multi-device playback

Built Dockerized real-time audio broadcasting system for synchronized playback across multiple devices using UDP multicast achieving 50-200ms latency. Integrated FFmpeg for AAC encoding, PulseAudio for audio capture, and lighttpd for web-based client distribution with QR code connectivity. Supports cross-platform clients (iOS, Android, Windows, macOS, Linux) with configurable bitrate (128k-320k) for network-adaptive streaming.